

Ryan Regalado

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

Bachelor of Science in Statistics, Minors in Computer Science & Mathematics

Aug. 2024 – Expected: May 2028

- **Achievements:** 3.93 GPA — Dean's List (Fall 2024, Spring 2025)
- **Relevant Coursework:** Optimization for Data Science, Nonlinear Optimization, Stochastic Processes, Statistical Machine Learning, Regression Analysis, Mathematical Statistics I & II

EXPERIENCE

Computational and Neurodynamics (CONE) Lab

Aug. 2026 – Present

Undergraduate Researcher

Urbana, IL

- Developing latent-position models in PyTorch to identify shared low-dimensional structure across longitudinal brain networks using Gaussian kernels and nonlinear optimization.

UIUC Department of Statistics

Jun. 2026 – Aug. 2026

Blackwell Research Scholar

Champaign, IL

- Developed a Python simulation framework to evaluate a high-dimensional statistical model across systematically varied experimental conditions.
- Engineered scalable linear algebra workflows using IRLB-based truncated SVD and parallel CPU computation, reducing estimation runtime by 66%.
- Applied the GJIVE model to 734-subject fMRI connectivity data with ~19K features, extracting interpretable representations of high-dimensional brain connectivity structure.
- Communicated research findings through interactive Streamlit dashboards, simulation visualizations, technical LaTeX reports, and documented Python code.

Illinois Mathematics Lab

Jan. 2026 – May 2026

Undergraduate Researcher

Champaign, IL

- Designed probabilistic metrics and simulation methods to identify anomalous performance patterns across 500,000+ NBA, MLB, and European soccer games.
- Co-developed a data engineering pipeline and database schema integrating 500,000+ games from multiple sources.
- Organized a collaborative repository of 300+ notebooks, scripts, datasets, and figures across the research team.
- Accelerated simulation-heavy outputs by 18x through vectorized computations.

TECHNICAL SKILLS

Languages: Python, R, SQL, C++, Java

ML & Scientific Computing: PyTorch, scikit-learn, NumPy, pandas, SciPy, tidyverse

Methods: Machine Learning, Statistical Modeling, Optimization, NLP, Simulation

Tools: Git, Streamlit, LaTeX, Matplotlib, ggplot2

PROJECTS

Hotel Price Prediction | *R, tidyverse, caret, glmnet, randomForest*

Apr. 2026 – May 2026

- Developed and evaluated eight regression and classification models—including linear regression, LASSO, KNN, bagging, random forest, LDA, QDA, and Naive Bayes—on 110,000+ hotel bookings with 31 predictors.
- Achieved a best test RMSE of 24.9 using Random Forest and 66.3% classification accuracy using LDA through comparative model evaluation.
- Engineered quantile-based ADR classes and applied PCA to mitigate multicollinearity for QDA and Naive Bayes models.

Fraud Call Detection | *Python, pandas, scikit-learn, NLTK, TextBlob*

Jan. 2025 – May 2025

- Built an NLP-based fraud call classifier to help identify scam voicemail transcripts for consumer protection.
- Developed NLP pipeline using Regex and TextBlob, preparing a corpus of 5000+ call transcripts for vectorization via Term Frequency-Inverse Document Frequency (TF-IDF).
- Trained a logistic regression model for fraud detection, accounting for a 9:1 class imbalance ratio, yielding 81% precision.
- Deployed classification model on iPhone voicemail transcripts, achieving 80% accuracy on real-world test data.